DATA SHEET

For a complete data sheet, please also download:

- The IC06 74HC/HCT/HCU/HCMOS Logic Family Specifications
- The IC06 74HC/HCT/HCU/HCMOS Logic Package Information
- The IC06 74HC/HCT/HCU/HCMOS Logic Package Outlines

74HC/HCT563

Octal D-type transparent latch; 3-state; inverting

Product specification
File under Integrated Circuits, IC06

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Philips Semiconductors

PHILIPS

Octal D-type transparent latch; 3-state; inverting

74HC/HCT563

FEATURES

- 3-state inverting outputs for bus oriented applications
- Inputs and outputs on opposite sides of package allowing easy interface with microprocessor
- Common 3-state output enable input
- Output capability: bus driver
- I_{CC} category: MSI

GENERAL DESCRIPTION

The 74HC/HCT563 are high-speed Si-gate CMOS devices and are pin compatible with low power Schottky TTL (LSTTL). They are specified in compliance with JEDEC standard no. 7A.

The 74HC/HCT563 are octal D-type transparent latches featuring separate D-type inputs for each latch and inverting 3-state outputs for bus oriented applications.
A latch enable (LE) input and an output enable ($\overline{OE}$) input are common to all latches.

The "563" is functionally identical to the "573", but has inverted outputs.

The "563" consists of eight D-type transparent latches with 3-state inverting outputs. The LE and $\overline{OE}$ are common to all latches.
When LE is HIGH, data at the D_n inputs enter the latches. In this condition the latches are transparent, i.e. a latch output will change state each time its corresponding D-input changes.

When LE is LOW the latches store the information that was present at the D-inputs a set-up time preceding the HIGH-to-LOW transition of LE.
When $\overline{OE}$ is LOW, the contents of the 8 latches are available at the outputs.
When $\overline{OE}$ is HIGH, the outputs go to the high impedance OFF-state.
Operation of the $\overline{OE}$ input does not affect the state of the latches.

QUICK REFERENCE DATA

GND = 0 V; T_{amb} = 25 °C; $t_r = t_f$ = 6 ns

SYMBOL	PARAMETER	CONDITIONS	TYPICAL		UNIT
			HC	HCT	
t_{PHL}/ t_{PLH}	propagation delay D_n, LE to $\overline{Q}_n$	C_L = 15 pF; V_{CC} = 5 V	14	16	ns
C_I	input capacitance		3.5	3.5	pF
C_{PD}	power dissipation capacitance per latch	notes 1 and 2	19	19	pF

Notes

1. C_{PD} is used to determine the dynamic power dissipation (P_D in μW):

   $P_D = C_{PD} \times V_{CC}^2 \times f_i + \sum (C_L \times V_{CC}^2 \times f_o)$ where:

   f_i = input frequency in MHz

   f_o = output frequency in MHz

   $\sum (C_L \times V_{CC}^2 \times f_o)$ = sum of outputs

   C_L = output load capacitance in pF

   V_{CC} = supply voltage in V

2. For HC the condition is V_I = GND to V_{CC}
   for HCT the condition is V_I = GND to V_{CC} – 1.5 V

ORDERING INFORMATION

See *"74HC/HCT/HCU/HCMOS Logic Package Information"*.

PIN DESCRIPTION

PIN NO.	SYMBOL	NAME AND FUNCTION
2, 3, 4, 5, 6, 7, 8, 9	D_0 to D_7	data inputs
11	LE	latch enable input (active HIGH)
1	$\overline{OE}$	3-state output enable input (active LOW)
10	GND	ground (0 V)
19, 18, 17, 16, 15, 14, 13, 12	$\overline{Q}_0$ to $\overline{Q}_7$	3-state latch outputs
20	V_{CC}	positive supply voltage

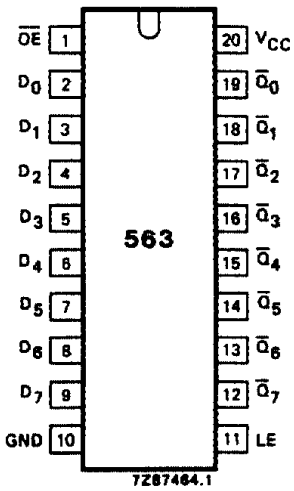


Fig.1 Pin configuration.

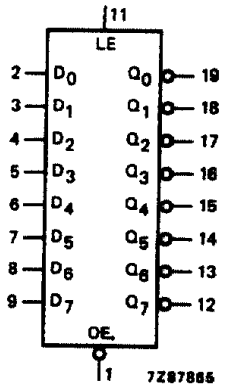


Fig.2 Logic symbol.

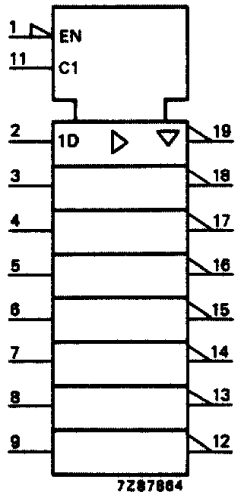


Fig.3 IEC logic symbol.

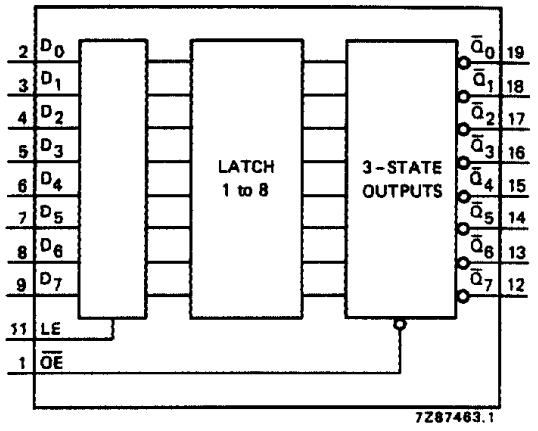


Fig.4 Functional diagram.

FUNCTION TABLE

OPERATING MODES	INPUTS			INTERNAL LATCHES	OUTPUTS
	$\overline{OE}$	LE	D_n		$\overline{Q}_0$ to $\overline{Q}_7$
enable and read register	L L	H H	L H	L H	H L
latch and read register	L L	L L	l h	L H	H L
latch register and disable outputs	H H	L L	l h	L H	Z Z

Notes

1. H = HIGH voltage level
   h = HIGH voltage level one set-up time prior to the HIGH-to-LOW LE transition
   L = LOW voltage level
   l = LOW voltage level one set-up time prior to the HIGH-to-LOW LE transition
   Z = high impedance OFF-state

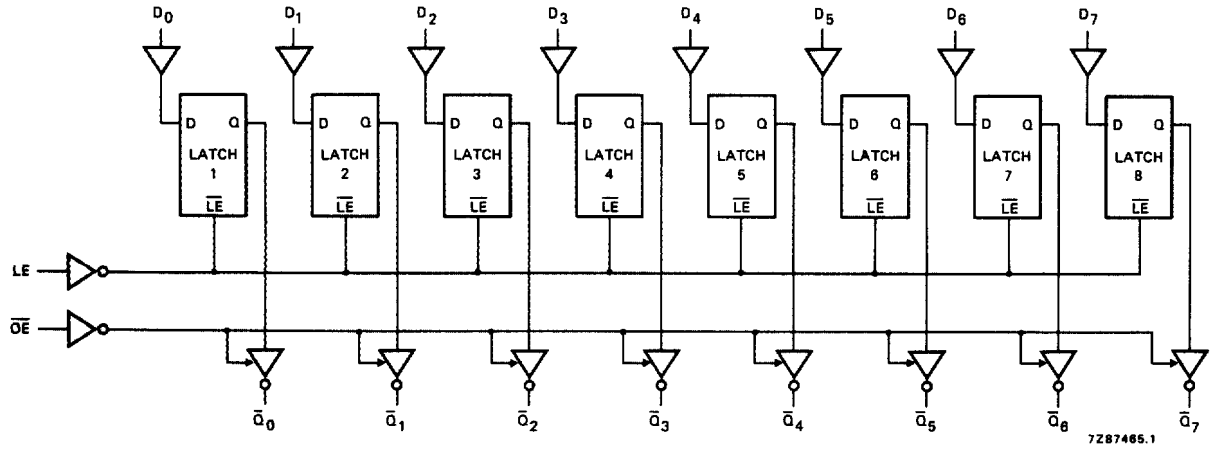


Fig.5 Logic diagram.

DC CHARACTERISTICS FOR 74HC

For the DC characteristics see *"74HC/HCT/HCU/HCMOS Logic Family Specifications"*.

Output capability: bus driver
I_{CC} category: MSI

AC CHARACTERISTICS FOR 74HC

GND = 0 V; $t_r = t_f$ = 6 ns; C_L = 50 pF

SYMBOL	PARAMETER	T_{amb} (°C)							UNIT	TEST CONDITIONS	
		74HC								V_{CC} (V)	WAVEFORMS
		+25			−40 to +85		−40 to +125				
		min.	typ.	max.	min.	max.	min.	max.			
t_{PHL}/ t_{PLH}	propagation delay D_n to $\overline{Q}_n$		47 17 14	145 29 25		180 36 31		220 44 38	ns	2.0 4.5 6.0	Fig.6
t_{PHL}/ t_{PLH}	propagation delay LE to $\overline{Q}_n$		47 17 14	145 29 25		180 36 31		220 44 38	ns	2.0 4.5 6.0	Fig.7
t_{PZH}/ t_{PZL}	3-state output enable time $\overline{OE}$ to $\overline{Q}_n$		47 17 14	150 30 26		190 38 33		225 45 38	ns	2.0 4.5 6.0	Fig.8
t_{PHZ}/ t_{PLZ}	3-state output disable time $\overline{OE}$ to $\overline{Q}_n$		50 18 14	150 30 26		190 38 33		225 45 38	ns	2.0 4.5 6.0	Fig.8
t_{THL}/ t_{TLH}	output transition time		14 5 4	60 12 10		75 15 13		90 18 15	ns	2.0 4.5 6.0	Fig.6
t_W	enable pulse width HIGH	80 16 14	14 5 4		100 20 17		120 24 20		ns	2.0 4.5 6.0	Fig.7
t_{su}	set-up time D_n to LE	50 10 9	11 4 3		65 13 11		75 15 13		ns	2.0 4.5 6.0	Fig.9
t_h	hold time D_n to LE	4 4 4	−6 −2 −2		4 4 4		4 4 4		ns	2.0 4.5 6.0	Fig.9

DC CHARACTERISTICS FOR 74HCT

For the DC characteristics see *"74HC/HCT/HCU/HCMOS Logic Family Specifications"*.

Output capability: bus driver
I_{CC} category: MSI

Note to HCT types

The value of additional quiescent supply current (ΔI_{CC}) for a unit load of 1 is given in the family specifications. To determine ΔI_{CC} per input, multiply this value by the unit load coefficient shown in the table below.

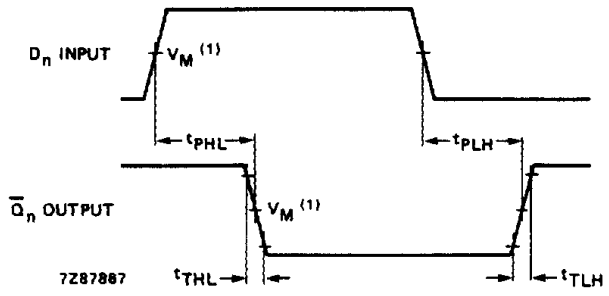
INPUT	UNIT LOAD COEFFICIENT
D_n	0.35
$\overline{LE}$	0.65
$\overline{OE}$	1.25

AC CHARACTERISTICS FOR 74HCT

GND = 0 V; $t_r = t_f$ = 6 ns; C_L = 50 pF

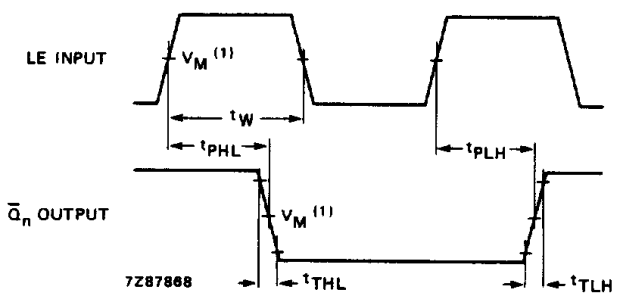
SYMBOL	PARAMETER	T_{amb} (°C)							UNIT	TEST CONDITIONS	
		74HCT								V_{CC} (V)	WAVEFORMS
		+25			−40 to +85		−40 to +125				
		min.	typ.	max.	min.	max.	min.	max.			
t_{PHL}/ t_{PLH}	propagation delay D_n to $\overline{Q}_n$		18	30		38		45	ns	4.5	Fig.6
t_{PHL}/ t_{PLH}	propagation delay LE to $\overline{Q}_n$		19	35		44		53	ns	4.5	Fig.7
t_{PZH}/ t_{PZL}	3-state output enable time $\overline{OE}$ to $\overline{Q}_n$		20	35		44		53	ns	4.5	Fig.8
t_{PHZ}/ t_{PLZ}	3-state output disable time $\overline{OE}$ to $\overline{Q}_n$		22	35		44		53	ns	4.5	Fig.8
t_{THL}/ t_{TLH}	output transition time		5	12		15		18	ns	4.5	Fig.6
t_W	enable pulse width HIGH	16	5		20		24		ns	4.5	Fig.7
t_{su}	set-up time D_n to LE	10	3		13		15		ns	4.5	Fig.9
t_h	hold time D_n to LE	5	−1		5		5		ns	4.5	Fig.9

AC WAVEFORMS



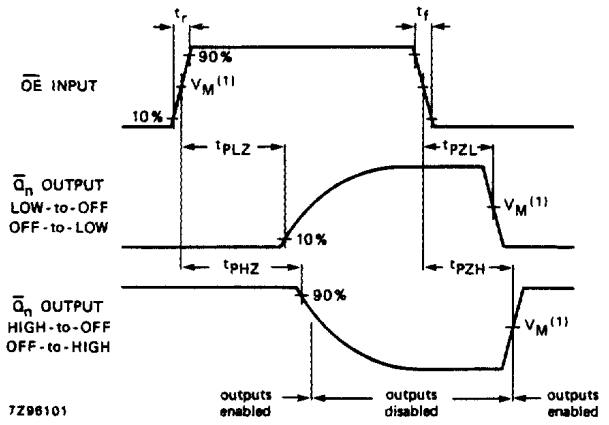
(1) HC : V_M = 50%; V_I = GND to V_{CC}.
HCT: V_M = 1.3 V; V_I = GND to 3 V.

Fig.6 Waveforms showing the data input (D_n) to output ($\overline{Q}_n$) propagation delays and the output transition times.



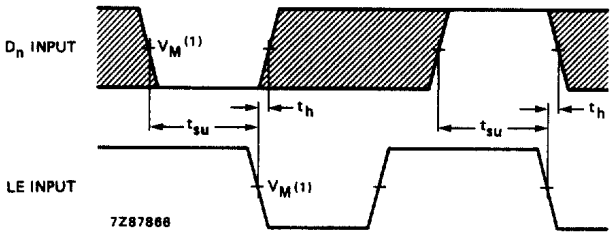
(1) HC : V_M = 50%; V_I = GND to V_{CC}.
HCT: V_M = 1.3 V; V_I = GND to 3 V.

Fig.7 Waveforms showing the latch enable input (LE) pulse width, the latch enable input to output ($\overline{Q}_n$) propagation delays and the output transition times.



(1) HC : V_M = 50%; V_I = GND to V_{CC}.
HCT: V_M = 1.3 V; V_I = GND to 3 V.

Fig.8 Waveforms showing the 3-state enable and disable times.



The shaded areas indicate when the input is permitted to change for predictable output performance.

(1) HC : V_M = 50%; V_I = GND to V_{CC}.
HCT: V_M = 1.3 V; V_I = GND to 3 V.

Fig.9 Waveforms showing the data set-up and hold times for D_n input to LE input

PACKAGE OUTLINES

See *"74HC/HCT/HCU/HCMOS Logic Package Outlines"*.